

(regular/irregular, large/small, etc), undetermined problems like those when there are resonating structures within the geometry or when the material contradicts in the problem.

Many advanced concepts, such as iterative solvers, preconditioned algorithms, domain decomposition methods and parallel computing, have been developed and implemented in present solvers.

CEM implementation has several disadvantages with the major one being the theoretical model results. The process is purely theoretical with no resemblance to experiments. So it would definitely need a validation with the practical results. Second, there will be numerical method error and mesh error that would go unnoticed, if not validated.

Numerical methods

The Maxwell equation can be written in two forms: integral and differential. Literally, integration means joining things and differentiation means dividing them. This literal definition of the two basically helps in determining which type of geometry can be applied with which type of Maxwell equation. If the geometry is planar and complex then the numerical method using integral Maxwell equation would be preferred.

The material boundaries in complex geometry cause discontinuities in the electric and magnetic fields. The type of Maxwell equations are then applied to these discontinuities, thus reducing the problem to algebraic expression (integral ones) and difference expressions (in case of differential Maxwell equation). Numerical methods employing differential Maxwell equation discretise the geometry by volumetric mesh (hexahedrons and tetrahedrons) and require an extra absorption boundary condition (ABC) type boundary over the geometry. In these methods the computations are done for the EM fields.

The numerical methods employing integral form of Maxwell equation discretise the geometry by surface mesh

(triangular, quads) and do not require extra boundary over the geometry. Computations are done for the current and charge although the 3D patterns can be generated.

What determines the choice of numerical method

Apart from the planar and 3D type geometry the numerical method is chosen on the basis of required accuracy of the result, the memory limitation of method, what different type of mate-

rial properties it can handle, whether the results have to be in time domain or in frequency domain, what would be its operating frequency, etc. These are secondary factors. The primary factor that determines the type of numerical method is the electrical size/length of the structure itself.

Electrical length is defined as ratio of physical length of device to the operating wavelength. As the electrical length increases or decreases, method to be used will also vary, as shown in Fig. 4.

Domains in which solvers work

Like time and frequency domains in the signal analysis, the CEM techniques are also domain-specific and are optimised for those. For CEMs there are basically five domains, namely, static, time, frequency, Eigen and ray tracing, as depicted in Fig. 5.

Static. In this domain there is no time variation of the equations and it is mostly used for DC bias field configuration of low field approximation.

Time. In this method the unknown field quantities are real-time varying in nature. This method does not use traditional matrix method, instead nearest neighbour interaction is calculated and interpolated. The applied source is impulsive in nature.

Frequency. This is a good choice if we need to calculate results all over the bandwidth of the centre frequency.

The excitation is sinusoidal in nature and broadband results are transformed into time domain by inverse Fourier transform. However, for low frequency this method is unsuitable, so in that case we need to use static domain methods. Also, each frequency needs separate computation, so frequency sweep should be selected appropriately. Otherwise, it would take a lot of memory.

Eigen. In this domain no excitation source is applied. This method is used to find

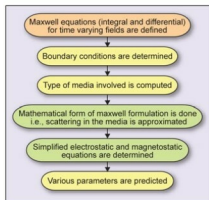


Fig. 3: General steps involved in a numerical method

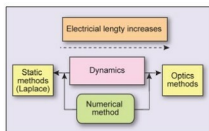


Fig. 4(a): Electrical length factor by which a particular type of method is used

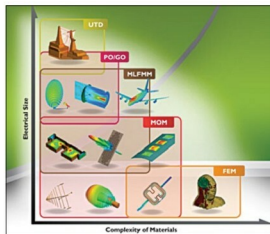


Fig. 4(b): Electrical size vs complexity of materials (Courtesy: FEKO-EM Simulation Software)